INCH-POUND

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DETAIL SPECIFICATION

# INFLATION DEVICE, AUTOMATIC, FLU-8B/P AND FLU-9B/P

*This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.*

## 1. SCOPE

**1.1 Scope.** This document gives the requirements for the manufacture, assembly, and packaging of the inflation device, automatic, FLU-8B/P and FLU-9B/P and the methods of inspection and tests upon which product acceptance will be based.

## 2. APPLICABLE DOCUMENTS

**2.1 General.** The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

**2.2 Government documents.**

**2.2.1 Specifications, standards, and handbooks.** The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Commander, Indian Head Division, Naval Surface Warfare Center, Technical Information Branch (Code 4230), 101 Strauss Avenue, Indian Head, Maryland 20640-5035, or emailed to amanda.penn@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

AMSC N/A FSC 1377

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STANDARDS

DEPARTMENT OF DEFENSE

| | |
|---|---|
| MIL-STD-1168 | Ammunition Lot Numbering and Ammunition Data Cards |
| DOD-STD-2101 | Classification Of Characteristics |

(Copies of these documents are available online at http://assist.daps.dla.mil/ or from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094)

**2.2.2 Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

Naval Air Systems Command

| | |
|---|---|
| 30003-849AS103 | Battery, Dry Storage |
| 30003-849AS153 | Inflation Device, Automatic FLU-8B/P |
| 30003-849AS221 | Inflation Device, Automatic FLU-9B/P |
| SPI 520-174-0272 | Packaging for Inflation Device |

(Application for copies should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Technical Information Branch (Code 4230), 101 Strauss Avenue, Indian Head, MD 20640-5035.)

**2.3 Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

American Society for Testing Materials

| | |
|---|---|
| ASTM E 1742 | Standard Practice for Radiographic Examination |

( Copies of this document are available online at http://www.astm.org or from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959)

American Society for Quality

| | |
|---|---|
| ASQ Z1.4 | Sampling Procedures and Tables for Inspection by Attributes |



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(Copies of this document are available online at http://www.asq.org/ or from the American Society for Quality, PO Box 3005, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)

**2.4 Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

**3.1 First article.** When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.2.

**3.2 General.** The FLU-8B/P and FLU-9B/P Inflation Devices shall be manufactured in accordance with Drawing 30003-849AS153, or Drawing 30003-849AS221 and all documents and drawings listed thereon. Lot numbers shall be in accordance with MIL-STD-1168.

**3.3 Product Baseline.** The manufacturer shall maintain a product baseline list. The list shall be complete, specifying each item used in the FLU-8B/P and FLU-9B/P. The list shall contain for each item: The Description, Part Number, or Identifying Number, Revision letter, and material and/or applicable specification. This list and subsequent revisions shall be submitted for technical review and shall be approved in writing by the Government Design Activity (Naval Surface Warfare Center, Indian Head, MD 20640-5035). Only FLU-8B/P and FLU-9B/P assemblies conforming to this specification and approved product baseline list will be accepted for delivery.

**3.4 Primary components.** For the purposes of this specification, the P/N CC-114-01 Primer is defined as a primary component (see 4.3.3).

**3.4.1 Primer P/N CC-114-01.** Prior to installation in the inflation devices each P/N CC-114-01 Primer shall be subjected to the following inspections.

**3.4.1.1 Bridgewire Resistance**. The bridgewire resistance of each primer shall be between 2.0 and 5.0 Ohms when tested per 4.6.1. The primer bridgewire resistance shall be measured prior to installation into the body assembly, and prior to installation of the sensor assembly.

**3.4.1.2 Radiographic Inspection.** The primers shall be free of imperfections when examined in accordance with 4.6.2.

**3.4.1.3 Ballistic Acceptance Tests.** The primer lot shall have been ballistically tested and accepted per the contractors acceptance procedure as approved by the Government Design Activity.

**3.5 Battery**. Batteries shall be per Drawing 30003-849AS103 and shall have been manufactured within 12 months of the delivery date. Batteries shall be included, but not installed, with inflators delivered for first article and lot acceptance test conformance inspection.

**3.6. Elastomeric Materials.** Elastomeric materials shall have been manufactured within 12 months of installation in the inflation device.



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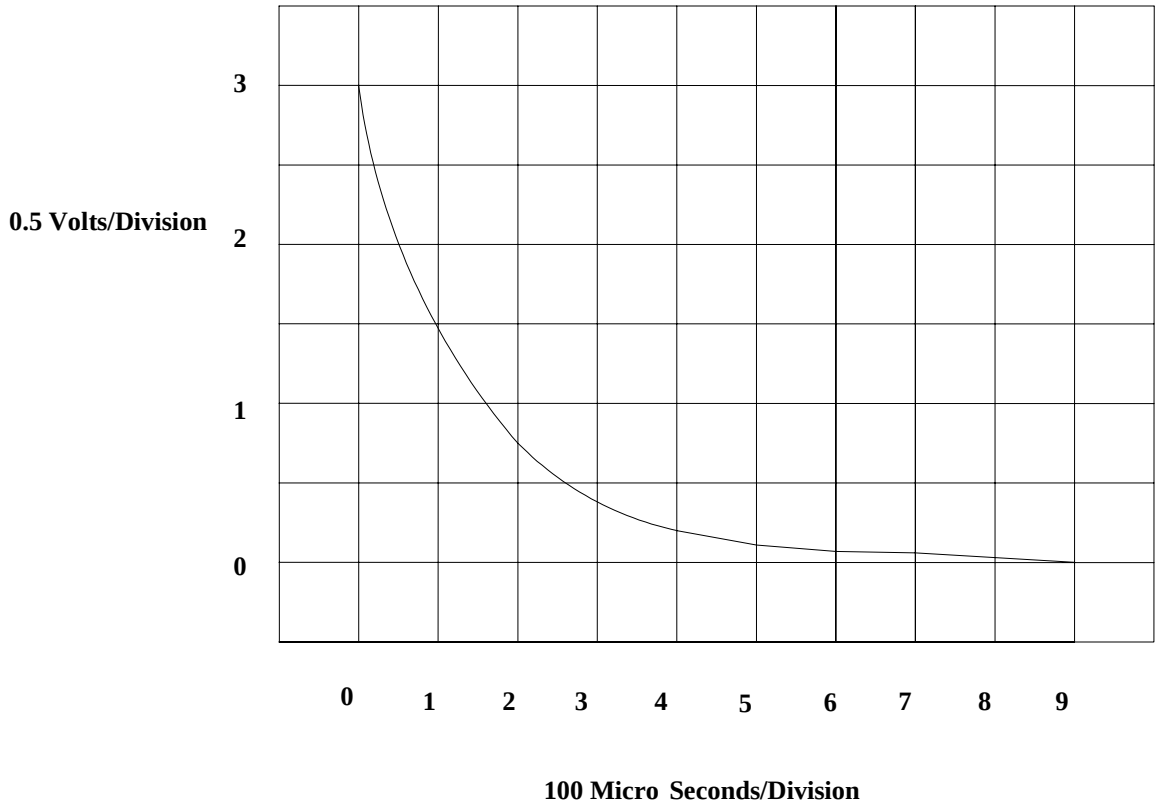



Figure 1. Sensor Assembly Voltage Output Curve

**3.7 Sensor Assembly Functional Test.** Each sensor assembly shall have an output voltage of 3.0 Volts minimum and have a voltage output curve similar to that shown in Figure 1 when tested per 4.6.3.

**3.8 Leakage.** Leakage tests shall be performed on inflation devices without shear wire or batteries installed.

**3.8.1 Leakage at 40 PSIG.** The inflation devices shall not show any evidence of leakage when tested per 4.6.9.1.

**3.8.2 Leakage at 10 PSIG.** The inflation devices shall not show any evidence of leakage when tested per 4.6.9.2.

**3.9 Manual Operation.**

**3.9.1 Lever Operation Without $CO_2$ Cylinder or Shear Wire.** Each inflation device shall fully stroke the piercing pin when tested per 4.6.10.1.



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**3.9.2 Lever Operation With $CO_2$ Cylinder.** Sample inflation devices shall fully stroke the piercing pin, perforate the $CO_2$ cylinder and release the $CO_2$ gas when tested per 4.6.10.2.

**3.10 Bleed Circuit Resistance**. Each inflation device shall have a bleed circuit resistance between 132,300 and 161,700 Ohms when tested per 4.6.11.

**3.11 Visual Examination.** Each inflation device shall conform to Drawing 30003-849AS153, or Drawing 30003-849AS221. Each inflation device shall be free of external damage; defects such as cracks, dents, corrosion, illegible or incomplete markings; missing or misassembled parts when inspected per 4.6.4.

**3.12 Radiographic Inspection.** The inflation devices shall be free of imperfections when examined in accordance with 4.6.5.

**3.13 Battery Voltage.** Batteries installed in inflation devices shall have a voltage of 12.0 Volts minimum when measured per 4.6.6. The voltage reading shall start at a high reading and gradually shift downward before final stabilization.

**3.14 Environmental Tests.**

**3.14.1 Temperate-Shock/Humidity/Altitude.** The inflation devices with batteries installed, shall not function, or otherwise be adversely affected when tested per 4.6.8.1.

**3.14.2 Vibration.** The inflation devices with batteries installed, shall not function, or otherwise be adversely affected when tested per 4.6.8.2.

**3.14.3 Rain.** The inflation devices with batteries installed, shall not function, or otherwise be adversely affected when tested per 4.6.8.3.

**3.14.4 No Fire Sensitivity**. The inflation devices with batteries installed, shall not function, or otherwise be adversely affected when tested per 4.6.8.4.

**3.15 Automatic Operation.** Inflation devices shall meet the following requirements when tested per 4.6.12.

**3.15.1 Function Time.** Function time is the time from immersion of the inflation device in water to start of release of $CO_2$ gas. The function time shall be less than five seconds per 4.6.12.1.

**3.15.2 Gas Release.** The $CO_2$ gas shall be fully released. The $CO_2$ gas shall not pass through any part of the inflation device other than the gas passages in the outlet port per 4.6.12.1.2.

**3.15.3 Piercing Pin.** The piercing pin of the FLU-8B/P shall fully stroke and remain in the fully stroked position per 4.6.12.1.3. The FLU-9B/P piercing pin shall fully stroke, and may not remain in the fully stroked position.

**3.15.4 Integrity.** No part of the inflation device shall rupture or fragment per 4.6.12.1.4.

**3.15.5 Post Firing Inspection.**



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**3.15.5.1 Packing Loop Release.** The packing loop of the FLU-8B/P shall be released and free to be removed from the inflation device per 4.6.12.1.5.

**3.15.5.2 Zipper Release.** The FLU-9B/P shear wire shall be severed, and the zipper release cover plate shall be freed per 4.6.12.1.6.

**3.16 Lanyard Release.** The FLU-9B/P lanyard shall not break or fail when a tensile load of 70 pounds is applied to the lanyard per 4.6.7.

**3.17 Workmanship.** Inflation devices shall be constructed and finished in a manner to assure compliance with all requirements of this specification and associated drawings. Particular attention should be directed to cracks, voids, foreign matter, missing or improperly assembled components, cracked, flaked or insufficient surface coatings, material defects, illegible or incorrect markings, corrosion, or mechanical defects such as burrs, sharp edges, and dents.

## 4. VERIFICATION

**4.1 Classification of inspections.**

a. First article inspection (see 4.3.2)
b. Quality conformance inspection (see 4.5)

**4.2 Sampling.** Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier shall be in accordance with the provisions of ASQ Z1.4 . Unless otherwise specified in the contract, order, or requisition, all samples for test purposes shall be provided at the expense of the contractor.

**4.3 Lots.**

**4.3.1 Inspection lots.** Inspection lot definition, formation, and size shall be in accordance with ASQ Z1.4 level S-1.

**4.3.2 First article inspection.** Before entering into quantity production, an acceptable first article sample of 35 FLU-8B/P or FLU-9B/P inflation devices, and 70 batteries shall be prepared. This sample, manufactured using the same methods and procedures proposed for the production lot, is for the purpose of determining that the contractor, prior to starting production, is capable of producing items which comply with the technical requirements of the contract. Thirty of these inflation devices shall be expended in the tests listed in Table I and five shall be retained for investigative purposes. A first article sample acceptable for environmental and functional testing shall be defined as a sample which has met the requirements of 3.11 through 3.17 of this specification. The contractor shall deliver the inflation devices to the activity designated by the contracting agency for test. First article sample acceptance shall consist of the tests shown in Table I in the order specified. Any quantity production prior to notification by the contracting agency of first article sample acceptability shall be at the contractor’s risk. The first article inflation devices will not be applied as a part of the quantity specified for delivery by the contract.



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**4.3.3 Production lot.** Lot numbers shall be in accordance with MIL-STD-1168. Inflation devices required for all test purposes are randomly selected from the production lot and will not be applied as a part of the quantity specified for delivery by the contract. Only primary components from a single lot shall be used in a production lot of inflation devices. One primary component production lot may be used in more than one inflation device production lot. The age of elastomeric materials shall be verified by inspection of the manufacturers manufacture date. Inflation devices selected for the lot acceptance test shall be delivered to the activity designated by the contracting agency. The manufacturer shall perform the tests and examinations listed in Table II. The designated test activity shall perform the tests and examinations listed in Table III to determine lot acceptance.

TABLE I. First Article Tests

| Test/ Inspection | Requirement Paragraph | Procedure Paragraph | Group A | Group B | Group C | Group D | Total |
|---|---|---|---|---|---|---|---|
| Visual | 3.11 | 4.6.4 | 18 | 6 | 6 | 5 | 35 |
| Radiographic | 3.12 | 4.6.5 | 18 | 6 | 6 | 5 | 35 |
| TSHA | 3.14.1 | 4.6.8.1 | 18 | | | | 18 |
| Vibration | 3.14.2 | 4.6.8.2 | | 6 | | | 6 |
| Rain | 3.14.3 | 4.6.8.3 | | | 6 | | 6 |
| Radiographic | 3.12 | 4.6.5 | | 6 | | | 6 |
| Battery Voltage | 3.13 | 4.6.6 | 18 | 6 | 6 | | 30 |
| No Fire Sensitivity | 3.14.4 | 4.6.8.4 | 18 | 6 | 6 | | 30 |
| Function at –20 F | 3.15 | 4.6.12.1 | 6 | 2 | 2 | | 10 |
| Function at 70 F | 3.15 | 4.6.12.1 | 6 | 2 | 2 | | 10 |
| Function at 120 F | 3.15 | 4.6.12.1 | 6 | 2 | 2 | | 10 |
| Retained Sample | | | | | | 5 | 5 |

**4.4 Classification of characteristics.** The characteristics verified by the tests and examinations herein are classified as Critical, Major or Minor in accordance with DOD-STD-2101. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

**4.5 Quality conformance inspection.** Quality conformance inspection shall consist of a lot inspection of all inflation devices per Table II and a sample lot acceptance per Table III. Any inflation device which does not meet the requirements of each test and examination in Table II shall be removed from the production lot. The lot acceptance test sample shall be per Table IV. Lot acceptance shall be per Table III in the order shown. Any failure shall reject the lot.

**4.6 Tests.**

**4.6.1 Bridgewire resistance.** The bridgewire resistance of 3.4.1.1. shall be measured with a test circuit which limits the bridge current to 5 milliamperes, maximum. A suitable instrument which is known to be accurate within one percent in the range of resistance specified shall be used to determine whether or not the requirement is met. Any primer not meeting requirements of 3.4.1.1 shall be rejected.

**4.6.2 Primer Radiographic Examination.** Each primer shall be radiographically examined in accordance with ASTM E 1742; any observable imperfections in assembly shall be cause for rejection. The primers shall be positioned on their sides for the most revealing exposure with the plane of the pins perpendicular to the X-ray source. All primers shall be identified utilizing a unique numbering system



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prior to examination. The primers shall be arranged sequentially on tray boards utilizing this system. Each radiographic plate shall carry a permanent means enabling identification of individual primers displayed there on. The radiograph identification shall include the complete lot number and the span of numbers displayed. Discontinuities in the numbering sequence shall be identified on the X-ray. Defective primers found by the vendor shall be clearly identified on the X-ray and those defectives removed from the production lot. Primers are to be left on trays or boards until after the X-rays have been reviewed and the defects identified to facilitate removal of defectives. The Vendor shall retain the primer radiographs.

TABLE II. Lot Inspections.

| Item | Nature of test | Requirement | Test Method | Classification (DOD-STD-2101) |
|---|---|---|---|---|
| Primary component | Single lot verification | 3.4 | 4.3.3 | (M101) |
| Primary component | Bridgewire Resistance | 3.4.1.1 | 4.6.1 | (M102) |
| Primary component | Radiographic Inspection | 3.4.1.2 | 4.6.2 | (M103) |
| Primary component | Ballistic Acceptance | 3.4.1.3 | | (C1) |
| Battery* | Age | 3.5 | 4.6.6 | (M104) |
| Elastomeric Materials | Age | 3.6 | 4.3.3 | (M105) |
| Sensor Assembly | Function | 3.7 | 4.6.3 | (C2) |
| FLU-9 Lanyard Assembly Test | Tensile Test | 3.6.1 | 4.6.7 | (M106) |
| Inflation Device | Leakage | 3.8 | 4.6.9 | (M107) |
| Inflation Device | Lever Operation w/o Cylinder | 3.9.1 | 4.6.10.1 | (M108) |
| Inflation Device | Lever Operation with Cylinder | 3.9.2 | 4.6.10.2 | (M109) |
| Inflation Device | Bleed Circuit Resistance | 3.10 | 4.6.11 | (M109) |
| Inflation Device | Visual Examination | 3.11 | 4.6.4 | (M110) |
| Inflation Device | Radiographic Inspection | 3.12 | 4.6.5 | (M111) |
| Inflation Device* | Battery Voltage | 3.12 | 4.6.6 | (M112) |

*Note: When batteries are supplied with production units.



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TABLE III. Sample Lot Acceptance Tests.

| Item | Nature of test | Requirement | Test Method | Classification (DOD-STD-2101) |
|---|---|---|---|---|
| Inflation Device | Radiographic Inspection | 3.12 | 4.6.5 | (M111) |
| Inflation Device | Visual Examination | 3.11 | 4.6.4 | (M110) |
| Inflation Device | Leakage | 3.8 | 4.6.9 | (M107) |
| Inflation Device | Battery Voltage | 3.12 | 4.6.6 | (M112) |
| Inflation Device | No Fire Sensitivity | 3.14.4 | 4.6.8.4 | (M113) |
| Inflation Device | Battery Voltage | 3.12 | 4.6.6 | (M112) |
| Inflation Device | Automatic Operation | 3.15 | 4.6.12 | (C3) |

TABLE IV. Sample selection

| Lot Size | Sample Size | Retain |
|---|---|---|
| 26 to 90 | 5 | 2 |
| 91 to 150 | 8 | 2 |
| 151 to 280 | 13 | 2 |
| 281 to 500 | 20 | 2 |
| 501 to 700 | 24 | 2 |
| 701 to 900 | 28 | 2 |
| 901 to 1200 | 32 | 2 |

**4.6.3 Sensor Assembly Functional Test.** Each sensor assembly shall be subjected to a simulated firing. The test fixture used shall provide a means of verifying the operation of the sensor assembly by applying a dummy resistance load to simulate the primer and an 18,000 Ohm (minimum) resistor to simulate the water resistance upon immersion. The output voltage wave form of each sensor shall be evaluated. Failure to meet the requirements of 3.7 shall be cause for rejection of the unit. A voltage output curve demonstrating a major discontinuity shall be cause for rejection.

**4.6.4 Visual Examination.** Inflation devices shall be visually examined and meet the requirements of 3.11.

**4.6.5 Radiographic Examination.** Each inflation device shall be radiographically examined in accordance with ASTM E 1742; any observable imperfections in assembly shall be cause for rejection. The inflation devices shall be positioned on their sides for the most revealing exposure with the plane of the lever perpendicular to the X-ray source. All inflation devices shall be identified with serial numbers prior to examination. The inflation device serial numbers shall be in consecutive order beginning with the



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lowest number in each production lot. The inflation devices shall be arranged on tray boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the inflation devices displayed thereon. The radiograph identification shall include the complete lot number, and the span of serial numbers displayed. Discontinuities in serial numbers shall be identified on the X-ray. Defective inflation devices found by the vendor shall be clearly identified on the X-ray and those defectives removed from the production lot. The vendor shall retain the inflation device radiographs.

**4.6.6 Battery.** Battery age shall be verified by inspection of manufacturers manufacture date. When required by drawing, batteries shall be installed in inflation devices for the battery voltage test. The voltage of each inflation device shall be measured using a volt meter or multimeter set on the volts DC scale and at the proper range. The minimum meter voltage range shall be 15 volts and the meter input impedance shall be 10 megohms minimum. The positive lead shall be attached to the sensor assembly mounting screw and the negative lead shall contact the sensor pin. The voltage reading shall start at a high reading and gradually shift downward before final stabilization. If no downward shift before stabilization occurs, the inflation device shall be rejected. Batteries that do not meet the requirements of 3.13 shall be replaced.

**4.6.7 Lanyard Tensile Test.** Ten percent of the FLU-9B/P inflators with lanyard installed shall be installed in a suitable tensile tester. The Lanyard shall be dry and not twisted more than one half turn. The tensile load shall slowly be applied until 70 pounds force is applied between the knob and the cam lever. The lanyards shall meet the requirements of 3.16.

**4.6.8 Environmental Tests.**

**4.6.8.1 Temperature-Shock/Humidity/Altitude.** The fully assembled inflation devices including batteries and $CO_2$ cylinder shall be supported on screen trays so that all areas are exposed to the prescribed atmospheric conditions at all times. The D port in the inflation devices shall be plugged to preclude intrusion of water into the devices. The cycling scheduling is as follows:

| | | |
|---|---|---|
| Monday | 0800 | Place test items in a chamber maintained at +70°F at 50 percent relative humidity (RH). |
| | 1200 | Raise chamber temperature to +120°F and the RH to 95 percent. The chamber temperature shall reach +120°F at 95 percent RH not later than 1300. |
| | 1600 | Remove test items from above chamber and immediately place in a chamber maintained at -20°F at a pressure altitude of 70,000 feet (0.65 psi). |
| Tuesday | 0800 | Remove test items from above chamber and immediately place in a chamber maintained at +70°F at 50 percent RH. |
| | 1200 | Remove test items from above chamber and immediately place in chamber maintained at -20°F at pressure altitude of 70,000 feet (0.65 psi). |
| | 1600 | Remove test items from above and immediately place in |



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| | | |
|---|---|---|
| | | a chamber maintained at 120°F at 95 percent RH. |
| Wednesday | 0800 | Reduce chamber temperature to +70°F at 50 percent RH. The chamber temperature shall reach +70°F at 50 percent RH not later than 0900. |
| | 1200 | Raise chamber temperature to +120°F at 95 percent RH. The chamber temperature shall reach +120°F at 95 percent RH not later than 1300. |
| | 1600 | Remove test items from above chamber and immediately place in a chamber maintained at -20°F at a pressure altitude of 70,000 feet (0.65 psi). |
| Thursday | 0800 | Remove test items from above chamber and immediately place in a chamber maintained at +70°F at 50 percent RH. |
| | 1200 | Remove test items from the above chamber and immediately place in a chamber maintained at -20°F at a pressure altitude of 70,000 feet (0.65 psi). |
| | 1600 | Remove test items from above chamber and immediately place in a chamber maintained at +120°F at 95 percent RH. |
| Friday | 0800 | Reduce chamber temperature to +70°F at 50 percent RH. The chamber temperature shall reach +70°F at 50 percent RH not later than 1300. |
| | 1200 | Raise chamber temperature to +120°F at 95 percent RH. The chamber temperature shall reach +120°F at 95 percent RH not later than 1300. |
| | 1600 | Remove test items from above chamber and immediately place in a chamber maintained at -20°F at standard ambient pressure. |

This schedule shall be followed for a total of 4 weeks (28 days) except that on the second and fourth weekends the soak time shall be from 1200 on Friday until 0800 on Monday at a temperature of +120°F at 95 percent RH. The inflation devices shall not fire or incur damage during the test.

**4.6.8.2 Vibration.** The inflation devices shall be equally divided for vibration at -20°F, 70°F, and 120°F. The inflation devices shall be mounted on a suitable fixture that allows vibration in 3 mutually perpendicular axes. In each axis a resonance survey shall be conducted and the resonance points noted. Resonance is the ratio of vibration output acceleration versus input acceleration of 2 or greater. A vibration dwell shall be conducted for 30 minutes at each of the 4 most severe resonance points. The dwell shall be peaked in case the resonant point shifts. After resonant dwell the inflation devices shall be vibration cycled for the time shown in Table V. The vibration cycling shall be sinusoidal vibration to the



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levels shown in Figure 2. The frequency shall be logarithmically swept from 5 Hz to 2000 Hz and return to 5 Hz in 20 minutes.

**4.6.8.3 Rain.** The fully assembled inflation devices including batteries and $CO_2$ cylinder shall be placed on plastic screen trays to allow air to circulate freely, water to drain easily, and exposing all surfaces to the environment. The D port in the inflation devices shall be plugged to preclude intrusion of water into the devices. The rainfall rate shall be adjusted to 4 ± 1 inches per hour and wind velocity to 40 ± 4 mph. The rain water conductivity shall be 350 ± 35 micromho/cm. These conditions shall be maintained for 30 minutes. The tray shall then be rotated 90 degrees and the test repeated. The tray shall be rotated and the test repeated until all 4 sides have been tested. The inflation devices shall then be dried at 120°F for 24 hours.

**4.6.8.4 No Fire Sensitivity.** The inflation devices shall be temperature conditioned at 70 ±5°F for 4 hours minimum. A 160 K Ohm ± 4 percent resistor shall be connected between the sensor pin and one of the sensor assembly mounting screws for 60 seconds minimum.

**4.6.9 Leakage.**

**4.6.9.1 Leakage at 40 PSIG.** A discharged $CO_2$ cylinder or equivalent test fitting shall be inserted in the inflation device less batteries. The lever shall be in the actuated position during this inspection. The actuated position is considered to be the position of the lever after the piercing pin has punctured the $CO_2$ cylinder. A pressure of 40 psig shall be applied at the outlet of the inflation device for 10 seconds minimum. While the pressure is applied, the inflation device shall be observed for leakage. Any detectable pressure drop shall be cause for rejection of the inflation device. Upon completion of this inspection the inflation device shall be subjected to the inspection of 4.6.8.2.

**4.6.9.2 Leakage at 10 PSIG.** A discharged $CO_2$ cylinder or equivalent test fitting shall be inserted in the inflation device less batteries. The lever shall be in the actuated position. A pressure of 10 psig shall be applied at the outlet of the inflation device for 10 seconds minimum. While the pressure is applied, the lever shall be slowly moved through at least one complete normal operation and the inflation device shall be observed for leakage. Any indication of leakage shall be cause for rejection of the inflation device.

**4.6.10 Manual Operation**

**4.6.10.1 Lever Operation Without $CO_2$ Cylinder.** A nylon thread shall not secure the cam lever of the FLU-8B/P, and a shear wire shall not be installed in the FLU-9B/P. The inflation device shall be oriented vertically with the $CO_2$ cylinder up. Attach a 3 + 0, -0.5 pounds weight to the free end of the lever and observe lever for complete rotation. Any inflation device that the lever does not completely rotate shall be rejected. This test shall be applied to 4 percent of the lot rounded up.



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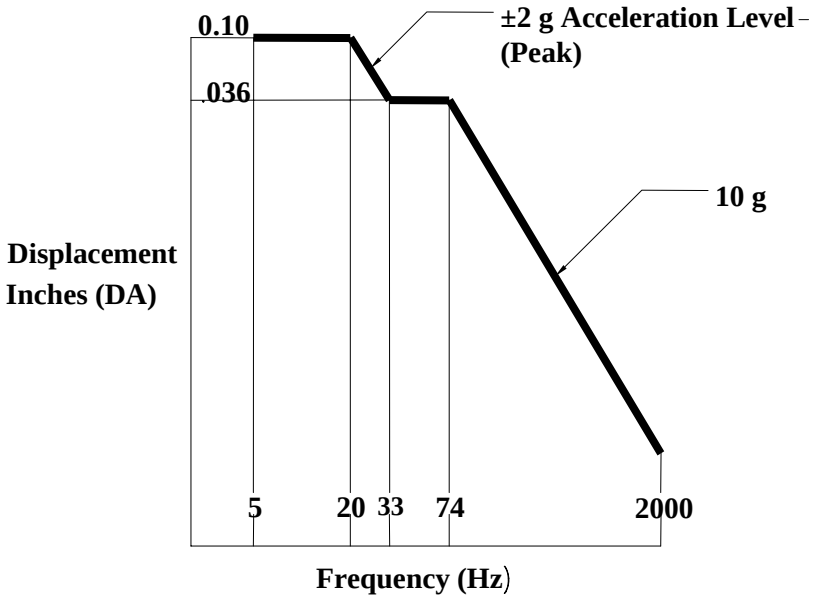



Figure 2. Vibration Test Levels

Table V. Vibration Test Schedule/Axis

| Resonances | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Total Dwell Time (minutes) | 0 | 30 | 60 | 90 | 120 |
| Total Cycling Time (minutes) | 180 | 150 | 120 | 90 | 60 |

**4.6.10.2 Lever Operation With $CO_2$ Cylinder.** The $CO_2$ filled cylinder shall be fully installed in the inflation device and hand tightened. For FLU-9B/P's the shear wire shall also be installed. The $CO_2$ cylinder shall then be removed and its diaphragm face examined for any evidence of puncture or indentations by the piercing pin. Upon completion of this inspection the cylinder shall be reinstalled in the inflation device. The inflation device shall be oriented vertically with the $CO_2$ cylinder up. Attach a 20 pound maximum weight to the free end of the lever. The piercing pin shall advance to its maximum point of travel, pierce the diaphragm of the $CO_2$ cylinder and release the $CO_2$ gas. Any inflation device that does not meet the requirements of 3.9.2 shall be rejected.

**4.6.11 Bleed Circuit Resistance**. The bleed circuit resistance shall be measured using a digital ohmmeter or a multimeter set on the ohms scale and the proper range. Equivalent equipment may be used to perform this test. The meter shall have a maximum 10 microamps output current and an open circuit voltage not exceeding 7.5 volts. The positive lead shall be attached to the positive battery contact and the negative lead to one of the sensor assembly mounting screws. The meter shall initially indicate a low resistance reading and shall count up (resistance reading increase) so that in approximately 45 seconds it stabilizes within the required resistance range. If the resistance reading is not per 3.10, or stabilizes immediately (no resistance increase) the inflation device shall be rejected.

**4.6.12 Automatic Operation**



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**4.6.12.1 Functional test.** The fully assembled inflation devices shall be equally divided for temperature conditioning at –20 ± 5°F, 70 ± 5°F, or 120 ± 5°F for a minimum of four hours. The inflation devices shall be functioned within three minutes after removal from temperature conditioning. The inflation devices shall be functioned by immersion in water at 32°F +5°F -0°F and 67 ± 5 micromho/centimeter conductivity. The time from immersion in water to start of release of $CO_2$ gas shall be measured.

**4.6.12.1.1 Acceptance criteria.** The total number of allowed defective units in a first article sample or lot acceptance sample as a result of tests and inspections of 3.8 through 3.15 shall be zero. Any defects will reject the lot represented. Screening of defects as a corrective action is at contractor's expense and the method of screening shall be subject to approval by the contracting officer.

**4.6.12.1.2 Gas Release.** The functioned inflation devices shall be inspected for complete release of gas from the $CO_2$ cylinder through the outlet port, see 3.9.2 and 3.15.2.

**4.6.12.1.3 Piercing Pin.** The functioned inflation devices shall be inspected for complete stroke of the piercing pin, see 3.15.3. Verification of full stroke of the FLU-9B/P piercing pin shall be by inspection of the $CO_2$ cylinder diaphragm.

**4.6.12.1.4 Integrity.** The functioned inflation devices shall be inspected for integrity, see 3.15.4.

**4.6.12.1.5 Packing Loop Release.** The functioned FLU-8B/P inflation devices shall be inspected for released packing loop, see 3.15.5.

**4.6.12.1.6 Zipper Release.** The functioned FLU-9B/P inflation devices shall be inspected for shear wire failure, and zipper release cover plate release.

**4.6.12.1.7 Retest.** There shall be no retests.

**4.6.12.1.8 Test failure.** If a test failure is attributable to an assignable cause, excluding the inflation devices, the original individual test results should be discarded and the complete test reconducted.

## 5. PACKAGING

**5.1 Packaging.** For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2 and 6.4). When packaging of materiel is to be performed by DOD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)



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**6.1 Intended use.** The FLU-8B/P and FLU-9B/P Inflation Devices are designed to inflate life preservers either manually or automatically by opening a $CO_2$ cylinder and releasing the gas into the life preserver through an outlet. These inflation devices were designed for military use and have no commercial application.

**6.2 Acquisition requirements.** Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.2.2).

c. Designation of inflation device (FLU-8B/P and FLU-9B/P)

d. Whether first article inspection is required and, if so, specify the test activity.

e. Production lot size and test activity (see 4.3.3).

f. Items of data required for each first article and production lot (see 6.3)

g. Packaging and marking (see 5.1 and 6.4).

h. That the safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition Ashore," OP 5 are applicable.

**6.3 Data.** For the information of contractors and contracting officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in section 2 of this specification or (c) referenced lower-tier documents need not be prepared for the Government unless specified in the contract or order.

**6.3.1 Standard deliverables.** In addition to the delivery of Inflation devices, the contract may require delivery of the following:

a. Radiographic plates for FLU-8B/P or FLU-9B/P inflators
b. Radiographic review report
c. Radiographic sample plates and techniques
d. Requests for waiver/deviation and engineering change proposal (ECP)
e. Test and inspection report including acceptance test plan if contractor tested
f. Ammunition data cards
g. Product Baseline
h. Failure analysis report and corrective action plan

**6.4 Contract packaging and marking.** The following packing and marking requirements should be specified in the contract.



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**6.4.1 Preservation and packing.** Preservation/packing should be in accordance with the Code of Federal Regulation 49 CFR 171-179. The contractor is be responsible for performance oriented packing certification and documentation in accordance with Regulation 49 CFR 171-179. Level A preservation and packing should be used for packaging of all production lots. Inflation devices should be packaged in an inner container in accordance with SPI 520-174-0272. Inner containers should be packed in the outer container in accordance with SPI 520-174-0272. For an explanation of terms see 6.4.2.

**6.4.2 Explanation of terms.** For the purpose of this specification, NAVSUPINST 4030.28 (20 June 1975) paragraphs 1-4, 2-2, 203 and Appendix A, when applicable, is used to clarify terms.

**6.4.3 Markings.**

**6.4.3.1 Special markings.** Marking of exterior containers should be in accordance with 49 CFR 171-179. Markings should be applied by the contractor or his agent.

**6.4.3.2 Normal markings.** Unless otherwise specified in the contractor order, the marking information on unit packages and shipping containers should be as specified on Drawing SPI 520-174-0272. The specified marking information should be applied to the containers in accordance with the applicable provisions of MIL-STD-129. Lot number should be in accordance with MIL-STD-1168.

**6.5 Definition.** Micromho/centimeter: The conductance in one centimeter of water through which one millionth (.000001) ampere flows when the potential is one volt.

**6.6 Subject term (keyword) listing.**

Inflator
Inflation Device

**6.4 Changes from previous issue.** Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.



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